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Machine Learning Using Multiparametric Magnetic Resonance Imaging Radiomic Feature Analysis to Predict Ki-67 in World Health Organization Grade I Meningiomas

BACKGROUND: Although World Health Organization (WHO) grade I meningiomas are considered “benign” tumors, an elevated Ki-67 is one crucial factor that has been shown to influence tumor behavior and clinical outcomes. The ability to preoperatively discern Ki-67 would confer the ability to guide surgical strategy.

OBJECTIVE: In this study, we develop a machine learning (ML) algorithm using radiomic feature analysis to predict Ki-67 in WHO grade I meningiomas.

METHODS: A retrospective analysis was performed for a cohort of 306 patients who underwent surgical resection of WHO grade I meningiomas. Preoperative magnetic resonance imaging was used to perform radiomic feature extraction followed by ML modeling using least absolute shrinkage and selection operator wrapped with support vector machine through nested cross-validation on a discovery cohort (n = 230), to stratify tumors based on Ki-67 <5% and ≥5%. The final model was independently tested on a replication cohort (n = 76).

RESULTS: An area under the receiver operating curve (AUC) of 0.84 (95% CI: 0.78-0.90) with a sensitivity of 84.1% and specificity of 73.3% was achieved in the discovery cohort. When this model was applied to the replication cohort, a similar high performance was achieved, with an AUC of 0.83 (95% CI: 0.73-0.94), sensitivity and specificity of 82.6% and 85.5%, respectively. The model demonstrated similar efficacy when applied to skull base and nonskull base tumors.

CONCLUSION: Our proposed radiomic feature analysis can be used to stratify WHO grade I meningiomas based on Ki-67 with excellent accuracy and can be applied to skull base and nonskull base tumors with similar performance achieved.

KEY WORDS: Artificial intelligence, Machine learning, Meningioma, Radiomics

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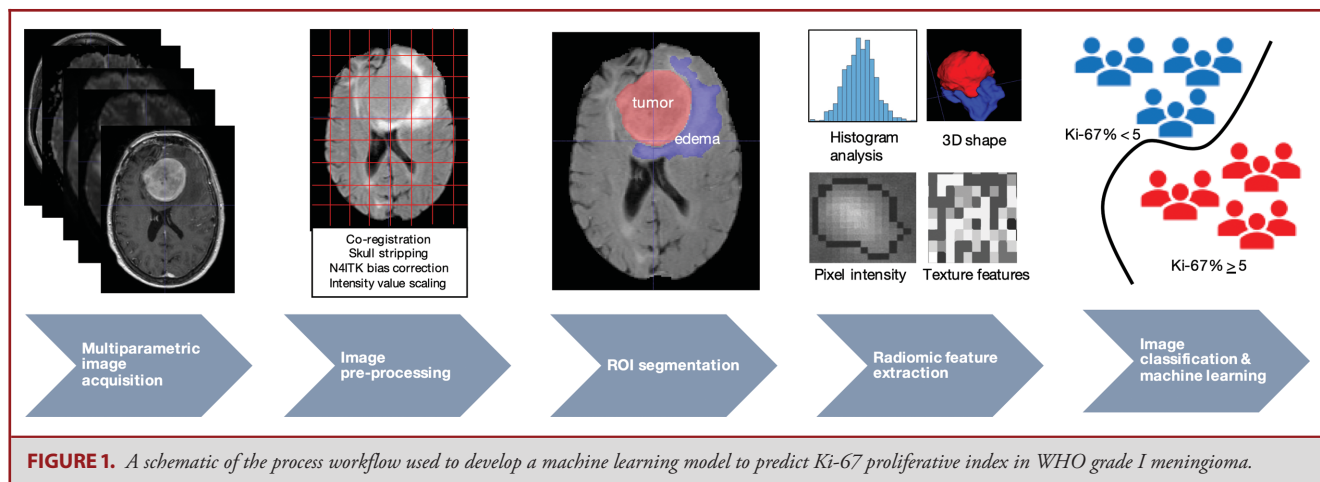
Under the 2016 WHO classification guidelines, grade I tumors account for 80% of all resected meningiomas, and are generally associated with favorable rates of local control with surgery alone.¹ However, up

to 30% of grade I tumors recur on long-term follow-up,² which is influenced by several factors, such as histopathology,^{3,4} degree of resection (Simpson grade),⁴⁻⁶ and Ki-67/MIB-1 proliferation index.^{7,8} As such, the ability to preoperatively prognosticate which tumors behave more aggressively would confer the ability to guide surgical strategy.

The Ki-67 proliferation index has been used as a surrogate marker of rapid growth and increased aggressiveness across a wide range of neoplasms.⁹⁻¹¹ Although its utility pertaining to meningioma remains controversial, several studies have shown that higher Ki-67 values correlate with increased risk of recurrence,¹²⁻¹⁴

ABBREVIATIONS: ADC, apparent diffusion coefficient; CV, cross-validation; LASSO, least absolute shrinkage and selection operator; ML, machine learning; MP, multiparametric; SVM, support vector machine; T1w, T1-weighted; T2w, T2-weighted; WHO, World Health Organization

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and serve as a predictor of time to recurrence.¹⁵ One recent study reported that in WHO grade I meningiomas with Ki-67% >4.5, patients with gross total resection incurred similar risk of recurrence compared to those with subtotal resection, and this cutoff represented an independent risk of recurrence regardless of extent of resection.⁴ Our institution's case series of 299 surgically resected WHO grade I meningiomas revealed that a Ki-67 of $\geq 5\%$ was associated with an increased risk of local recurrence.⁸ The current WHO guidelines do not utilize Ki-67 as a criteria for meningioma classification, despite the fact that there is a wide overlap of Ki-67 values that may be found between grade I, II, and III tumors, and, as such, the clinical behavior of meningiomas is not entirely captured by its designated grade, and there is a need to elucidate additional factors that portend increased aggressiveness.^{16,17}

Radiomic analysis utilizing multiparametric magnetic resonance imaging (MP-MRI) can be used to render high-throughput computational feature extraction, such as tumor size, shape, feature intensity, and texture patterns,¹⁸⁻²⁰ which can subsequently be studied by image classification machine learning algorithms to develop models that can predict tumor pathology^{21,22} and model outcomes,²³ all of which serve to optimize patient-specific treatment paradigms. Previous studies have investigated using machine learning to predict meningioma grade,^{24,25} with particular emphasis placed on discerning between low and high-grade tumors.^{26,27} Morin et al developed a complex model incorporating clinical, radiologic, and radiomic features to stratify meningiomas based on predicted clinical outcomes independent of tumor grade.²⁸ There have been no studies to date, however, which apply radiomics-based machine learning techniques to assess individual tumor biomarkers that may help prognosticate the heterogeneous clinical behavior amongst grade I meningiomas, which represent the vast majority of tumors encountered in clinical practice.

In this study, we use standard preoperative MP-MRI to perform radiomic feature extraction and develop a machine learning model to predict the Ki-67 proliferative index in WHO

grade I meningiomas. This is the largest meningioma radiomics machine learning study to date, and the only one to focus exclusively on grade I tumors, which have been shown to have greater molecular and radiographic homogeneity than grade II/III tumors, and provides the best opportunity to correlate sampled tumors' Ki-67 seen on pathology as a surrogate for the entire lesion seen on MRI.²⁹ The ability to predict Ki-67 preoperatively provides the surgeon with a simple yet important parameter that can be used to guide surgical strategy.

METHODS

Source of Data

The study protocol was approved by our Institutional Review Board, under a waiver of informed consent. A retrospective analysis was performed of all patients ($n = 376$) who underwent surgical resection of WHO grade I meningioma at our institution between 2012 and 2018. Pathology reports were reviewed and patients who met criteria for of grade I meningioma using the 2016 World Health Organization Classification of Tumors of the Central Nervous System were included in the present study.³⁰ Patients' electronic medical records were accessed, and relevant clinical data was collected. Eligible patients' surgical pathology reports were reviewed, and the Ki-67 proliferative index was recorded. In all patients, the Ki-67 was quantified using Aperio IHC image analysis software (Aperio Technologies Inc, Cumming, Georgia).

Preoperative MRI studies performed at our hospital on 3 different scanners, as well as from outside hospital prior to transfer to our institution, closest to the date of surgery were included in the study. MP-MRI data comprising of 7 total sequences: T1-weighted (T1w), volumetric T1-weighted postcontrast (T1 + C), T2-weighted (T2w), T2-fluid-attenuated inversion recovery (T2-FLAIR), diffusion weighted image (DWI) including b0 and b1000 images, and apparent diffusion coefficient (ADC) map were exported to a secure server storage system (MIM Software Inc, Beachwood, Ohio). Patients who did not have all 7 scans, as well as those with significant motion artifact, were excluded from the study. A total of 306 patients met inclusion criteria for the present study, and 70 were excluded due to limited data.

Image Preprocessing and Segmentation

A schematic showing the process of image processing and machine learning analysis is shown in Figure 1. Patients' MRI scans were anonymized, and all protected health information was stripped from DICOM files using automated software (DICOMAnon, Red Ion LLC, Birmingham, Alabama). Image preprocessing was performed using CaPTk v.1.8.1 (<https://www.med.upenn.edu/cbica/captk/>), an open-source, multiplatform software.^{31,32} Image preprocessing for each of the patients included coregistration of all MRI sequences upon its corresponding axial thin-cut (0.5 mm) T1 + C sequence, resampling of the images to $1 \times 1 \times 1 \text{ mm}^3$ spatial resolution, and N4ITK bias correction. The intensity values for all image volumes were scaled in the range of [0, 255] after removing pixels with outlier values.

Images were segmented using ITK-SNAP v3.8.0 (www.itksnap.org) by a neurosurgeon blinded to patients' clinical information and Ki-67 proliferation index. The enhancing tumor was segmented on the T1 + C sequence, and the peritumoral edema was identified on the T2-FLAIR sequence, although the segmentation schemes were overlaid on all 7 coregistered sequences for subsequent radiomic analysis.

Radiomic Analysis

For each patient, feature extraction was performed in CaPTk over the segmentation labels. Morphologic volume and shape features ($n = 29$) were computed from enhancing tumor and peritumoral edema regions. Radiomic features ($n = 2520$) pertaining to volume, shape, size, histogram, and texture parameters were extracted from MP-MRI scans,^{33,34} and were normalized using z-scoring prior to machine learning analysis.

A multivariate pattern classification algorithm was developed to stratify Ki-67 proliferative index of $<5\%$ and $\geq 5\%$.⁸ The dataset ($n = 306$) was split into discovery and independent replication cohorts (75% and 25% of the entire cohort, respectively). Model training and selection was carried out in the discovery cohort ($n = 230$). The replication cohort ($n = 76$) was kept "unseen" during the model training and was merely used as an independent set to provide an insight about the generalizability of our predictive model for prospective patients.

We trained our model based on the extracted radiomic features in the discovery cohort by the least absolute shrinkage and selection operator (LASSO) feature selection method, wrapped with linear support vector machine (SVM) classification through a nested cross-validation (CV) approach, with 10-fold CV for the outer and inner loops. SVMs are well suited to provide robust dichotomization of datasets by providing optimal hyperplanes which, in turn, may be effectively applied to an external "unseen" dataset. Furthermore, SVMs are computationally efficient. Other algorithms such as k -NN and random forest were considered but ultimately not selected given their penchant for overfitting of the high dimensionality training dataset that limits their generalizability. An internal 3-fold CV was set for the SVM model to search for the optimal value of the parameter for the soft margin cost function (C), through a Bayesian optimization approach. Among the 10 models trained in the outer folds, the model with the least overfitting and closest performance to the average performance of all 10 models was chosen. The LASSO method is well suited to reduce the likelihood of overfitting by assigning a penalty to over-represented feature coefficients in the model, and given that colinearity was not

TABLE 1. Baseline Demographic and Clinical Characteristics of the Included $n = 306$ Patients Who Underwent Surgical Resection of WHO Grade I Meningioma

Gender	
Male	99 (32.4%)
Female	207 (67.6%)
Mean age (range)	59 ± 14 (19–90)
Laterality of tumor	
Right	137 (44.8%)
Left	155 (50.6%)
Midline	14 (4.6%)
Skull base meningioma	
Clinoidal	7 (2.3%)
Olfactory groove	7 (2.3%)
Petroclival	8 (2.6%)
Tuberculum sellae/planum sphenoidale	5 (1.6%)
Sphenoid wing	44 (14.4%)
Nonskull base meningioma	
Convexity	79 (25.8%)
Intraventricular	4 (1.3%)
Parafalcine	14 (4.6%)
Parasagittal	79 (25.8%)
Pineal	1 (0.3%)
Posterior fossa	36 (11.8%)
Temporal/middle fossa	14 (4.6%)
Tentorium	8 (2.6%)
Ki-67 mean (range)	
Skull base meningioma	4.97 ± 3.54 (1–17)
Nonskull base meningioma	4.80 ± 4.17 (0.3–33.6)
Ki-67 median (Q1, Q3)	
Skull base meningioma	3.7 (2.59, 6.60)
Nonskull base meningioma	3.7 (2.18, 5.85)
Ki-67 distribution	
$<5\%$	214 patients
$\geq 5\%$	92 patients

felt to be an issue, ridge regression or elastic net was not pursued. This model was applied in the replication cohort to obtain the classification performance in independent patient data. The predictive modeling approach was implemented in MATLAB R2018b (Mathworks, Natick, Massachusetts).

Statistical Analysis

Frequencies and percentages are used for nominal variables, and means and ranges are used for continuous variables. Analysis was carried out using unpaired 2-tailed t -test, Chi-square, and Fisher's exact tests, as appropriate. Statistical analysis was carried out with IBM SPSS v 24.0 (IBM Corp, Armonk, New York) and Matlab 2018b.

RESULTS

Patient demographics and baseline clinical characteristics are summarized in Table 1. The mean patient age was 59 ± 14 yr (range: 19–90), and 67.6% of the study cohort was female. A total of 71 patients presented with skull base meningioma, and

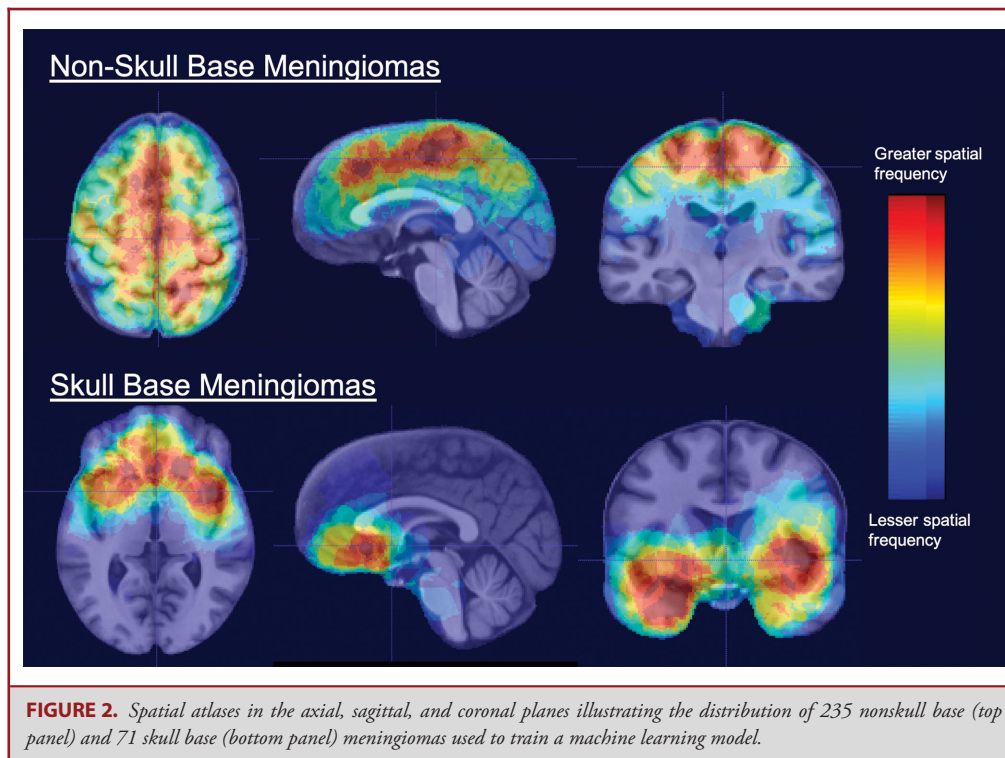


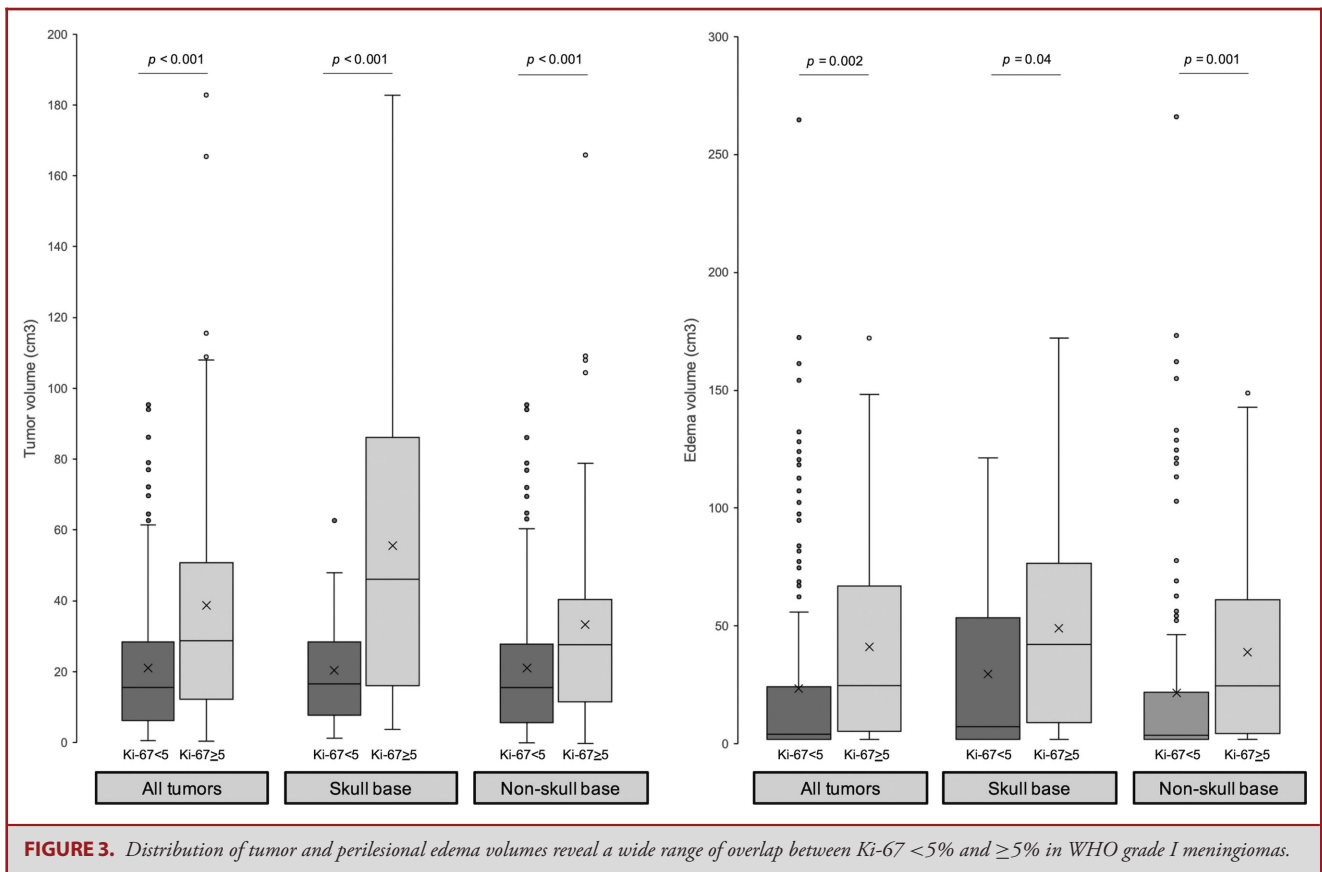
FIGURE 2. Spatial atlases in the axial, sagittal, and coronal planes illustrating the distribution of 235 nonskull base (top panel) and 71 skull base (bottom panel) meningiomas used to train a machine learning model.

the other 235 patients presented with nonskull base meningioma (Figure 2). The most common locations of tumors are convexity (25.8%), parasagittal (25.8%), and sphenoid wing (14.4%). The mean and median Ki-67 of tumor specimens were $4.84 \pm 4.03\%$ (range: 0.3-33.6) and 3.7% (Q1:2.3%, Q3:6%), respectively. There were 214 patients with Ki-67 <5% and 92 patients with Ki-67 of $\geq 5\%$. The mean Ki-67 values of the discovery ($n = 230$) and replication ($n = 76$) cohort were 4.8 ± 3.76 and 5.03 ± 4.78 , respectively ($P = .642$). A total of 71 subjects (30.87%) and 25 subjects (32.89%) had Ki-67 >5% in the discovery and replication cohort, respectively.

Morphologic volumetric analysis pertaining to tumor size and perilesional edema was performed using values obtained from feature extraction. There is wide overlap between both tumor and edema volumes between lesions with Ki-67 <5% and those $\geq 5\%$ (Figure 3). Meningiomas with Ki-67 $\geq 5\%$ were larger in volume compared to tumors with Ki-67 <5% (mean 38.65 ± 19.19 and $20.97 \pm 35.57 \text{ cm}^3$, respectively; $P < .001$), which held true in subgroup analysis of both skull base and nonskull base tumors. Similarly, meningiomas with Ki-67 $\geq 5\%$ had significantly larger peritumoral edema volumes compared to tumors with Ki-67 <5% (mean 22.11 ± 40.12 and $40.16 \pm 42.34 \text{ cm}^3$, respectively; $P = .002$), and this difference was more pronounced in nonskull base compared to skull-base tumors.

Following high-throughput radiomic feature extraction from MP-MRI, model training, including feature subset selection and hyperparameter tuning, was performed in the discovery cohort, resulting in a total of 60 radiomic features in the predictive model (Supplemental Table 1). With this model, an AUC of 0.84 (95% CI: 0.78-0.90), with associated sensitivity and specificity of 84.1% and 73.3%, respectively, was achieved in the discovery cohort. The selected features in the trained predictive model were then calculated for the subjects of the replication cohort and the model was applied independently in this cohort. An AUC of 0.83 (95% CI: 0.73-0.94), similar to the training discovery set, with a sensitivity of 82.6% and specificity of 85.5%, was obtained for this independent testing (Figure 4). Furthermore, the model performed commendably when applied to all skull base and nonskull base tumors in our patient cohort, evidenced by comparable AUC values of 0.86 and 0.83, respectively (Table 2).

All sequences of MRI contributed towards the final features list: DWI (b0 and b1000 images, and ADC map) produced the most features (27 total), followed by T1 + C (13). Furthermore, morphologic features such as increased peritumoral edema shape eccentricity and enhancing tumor extent were highly discriminative for tumors with Ki-67 $\geq 5\%$ versus those with Ki-67 <5% (Table 3). Figure 5 illustrates examples of 2 patients with sphenoid wing meningioma, one with a high (17%) and another with a low



(2.8%) Ki-67, and how quantitative radiomic features can be used to discriminate between the two.

DISCUSSION

Although the vast majority of meningiomas are classified as WHO grade I tumors, there is wide heterogeneity in their clinical presentation, rate of growth, and risk of recurrence.³⁵ Machine learning using radiomic feature analysis derived from standard anatomic MR images may reveal prognostic insights to predict tumor behavior. In this study, we have trained a machine learning model that can be used to predict Ki-67 in grade I meningiomas and help guide surgical timing and operative strategy. Patients who are known to harbor tumors with Ki-67 ≥5% should undergo more aggressive operative intervention with an earnest attempt at achieving a gross total resection with wide dural margins.

This study identifies several imaging characteristics that highlight morphologic differences between tumors in our dataset. Tumors with Ki-67 ≥5% were significantly larger than those with Ki-67 <5%, and this was true in both the skull base

and nonskull base cohorts. However, there was a wide overlap in tumor volume despite Ki-67 indices, and our model was able to effectively discern smaller tumors that harbored higher Ki-67 values. Similarly, the volume of peritumoral edema was greater in meningiomas with Ki-67 ≥5%, which corroborates a long-held belief that rapidly growing tumors—for which Ki-67 is a surrogate marker—exhibit greater degree of peritumoral edema.^{36,37} Morphologic features such as tumor extent and edema shape eccentricity were important predictive features that appeared in our model. Furthermore, radiomic features extracted from DWI and T1 + C sequences contributed the highest number of selected features.^{29,38} Taken together, these morphologic and radiomic findings highlight the benefits of using multiparametric MRI sequences instead of solely T1 + C sequence to improve the performance of machine learning models.

Previous studies that have applied machine learning techniques to meningiomas have focused on identifying tumors that are of higher grade,^{24,27} such that surgery can be tailored to achieve maximal resection, given the propensity of grade II and II tumors to exhibit recurrence despite adjuvant radiation treatment.²⁸ These studies have incorporated both adjunctive

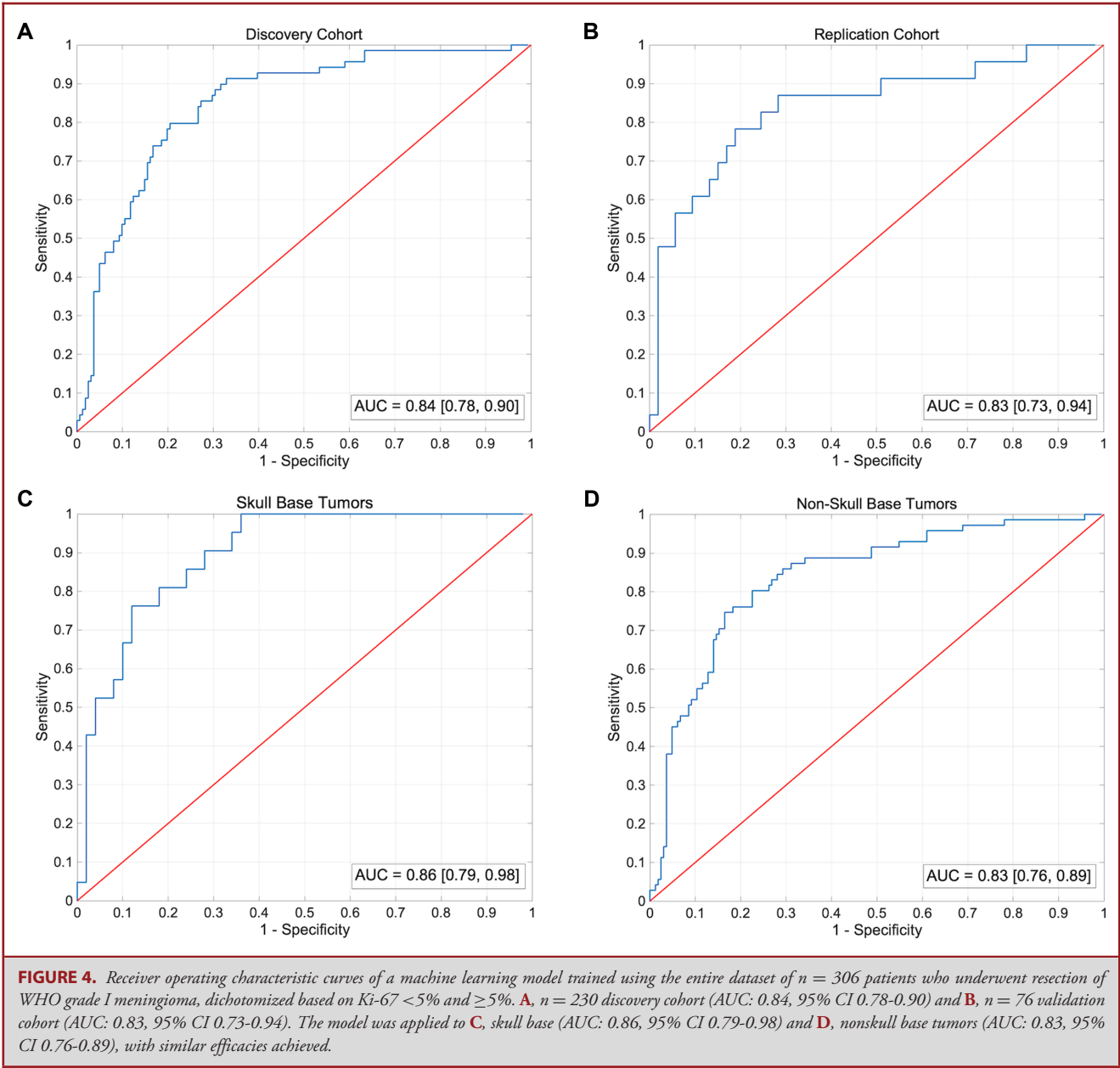


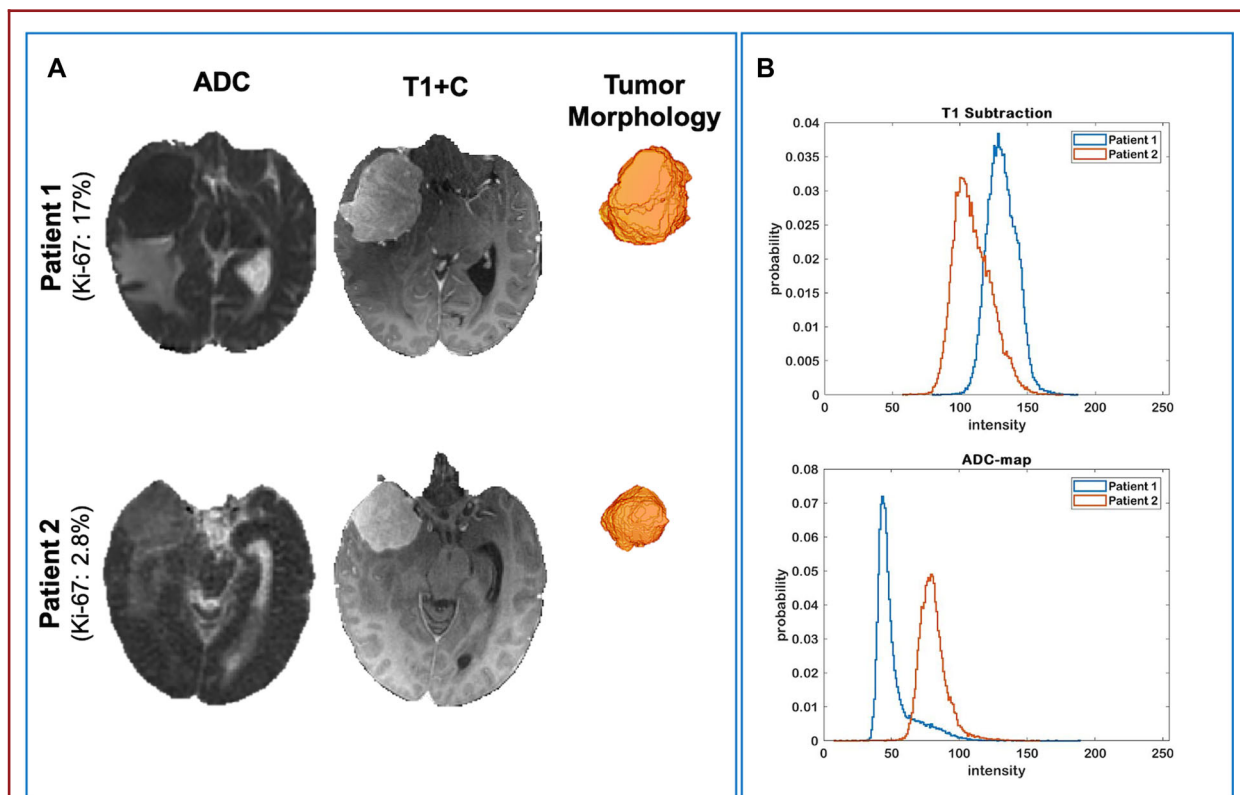
TABLE 2. A Summary of Performances of the Predictive Model in Classifying WHO Grade I Meningiomas Based on Ki-67 $<5\%$ and $\geq 5\%$ in Discovery ($n = 230$) and Replication ($n = 76$) Cohorts

	AUC [95% CI]	Sensitivity (%)	Specificity (%)
Discovery cohort	0.84 [0.78, 0.90]	84.1	73.3
Replication cohort	0.83 [0.73, 0.94]	82.6	85.5
Skull base tumors	0.86 [0.79, 0.98]	80.9	82.0
Nonskull base tumors	0.83 [0.76, 0.89]	80.3	77.4

The trained model performed similarly when applied to skull base ($n = 71$) and nonskull base ($n = 235$) tumors. AUC = area under the curve; CI = confidence interval.

TABLE 3. MP-MRI Was Used for Radiomic Model Training, Yielding a Total of $n = 60$ Features in the Final Predictive Model

MRI sequence	# of selected features	Selected feature categories
Morphological features	4	Edema shape eccentricity, tumor shape (extent), tumor shape (eccentricity)
T1-weighted	4	Histogram ($n = 1$), tumor intensity ($n = 1$), GLRLM ($n = 1$), GLSZM ($n = 1$)
T1-contrast enhanced	13	CoLIAGe ($n = 3$), Gabor ($n = 4$), GLSZM ($n = 1$), histogram ($n = 3$), intensity ($n = 1$), LBP ($n = 1$)
T2-weighted	3	Gabor ($n = 2$), histogram ($n = 1$)
T2-FLAIR	9	CoLIAGe ($n = 2$), Gabor ($n = 2$), GLCM ($n = 1$), GLSZM ($n = 1$), histogram ($n = 2$), LBP ($n = 1$)
DWI (b0)	8	CoLIAGe ($n = 2$), Gabor ($n = 2$), GLSZM ($n = 1$), histogram ($n = 3$)
DWI (b1000)	7	CoLIAGe ($n = 2$), Gabor ($n = 1$), GLCM ($n = 1$), histogram ($n = 2$), intensity ($n = 1$)
ADC map	12	Collage ($n = 4$), Gabor ($n = 2$), GLSZM ($n = 1$), histogram ($n = 5$)

**FIGURE 5.** Illustrative cases showcasing the radiomic phenotypes of 2 patients with sphenoid wing meningioma: patient 1 with Ki-67 of 17% and patient 2 with Ki-67 of 2.8%. **A**, T1 + C image shows similar hyperintensity albeit ADC maps with lower ADC values in the tumor with high Ki-67 compared to the tumor with lower Ki-67. **B**, Normalized histograms of T1-subtraction (T1 + C minus T1 intensity) and ADC map between patients, showing an association of higher T1-subtraction intensity (ie, increased permeability), and lower ADC values (ie, restricted diffusion or increased cellular density) within the tumorous region of with higher Ki-67.

clinical and qualitative radiographic variables that are input into the machine learning model, which require human interpretation and manual entry into the algorithm. Our study relies solely on radiomic features obtained from standardized preoperative MRI, which presents the possibility that this process may be fully automated and readily incorporated into clinical practice.³⁹ In this regard, our algorithm would serve to identify grade I

tumors whose behavior is more akin to grade II tumors, and therefore would influence frequency of surveillance and surgical strategy.

The ability to rule out high-grade meningiomas does not, in and of itself, eliminate all tumors that exhibit more aggressive behavior. In fact, there is a sizeable subset of grade I meningiomas that have a faster rate of growth and an increased risk

of recurrence, which is not entirely accounted for by the current WHO classification. Recently published studies have identified elevated Ki-67 as an important risk factor for recurrence in grade I meningioma, even in cases with reported gross total resection.^{4,8} The Simpson grade of resection as well as the volume of residual tumor has been shown to influence risk of recurrence,⁴⁰ which underscores the need to achieve a supramarginal resection whenever possible in patients with elevated Ki-67. Furthermore, the ability to predict the Ki-67 value could be invaluable in guiding the time interval for imaging surveillance and deciding when to pursue surgical intervention. For example, given their increased risk of expected tumor growth, patients identified who harbor tumors with elevated Ki-67 may be counseled to undergo surgical resection earlier to minimize surgical morbidity and maximize extent of resection (eg, parasagittal meningioma that has not yet grown to involve the superior sagittal sinus). Given the lack of available knowledge that accounts for their disparate clinical behavior, the incorporation of machine learning techniques would provide an important adjunct in the management of WHO grade I meningiomas.

Limitations

Our study is limited by its retrospective design, and an inherent selection bias. The Ki-67 values were not subject to re-review for the purposes of this study. Although there can be interobserver variability between reported Ki-67 values, all our specimens were analyzed via semi-automated image analysis software, which has been shown to improve the reproducibility of outcomes.^{41,42} Furthermore, the Ki-67 in grade I meningiomas has been shown to have less intratumoral heterogeneity compared to higher grade tumors.²⁹ Thorough validation of this machine learning model will ultimately require application to an external, multi-institutional dataset with a larger cohort of patients.

CONCLUSION

Machine learning using radiomic feature analysis is uniquely suited to identify phenotypic imaging signatures that can be used to offer enhanced tumor diagnostics. In this study, we have trained a machine learning model that is effective in stratifying WHO grade I meningiomas based on Ki-67 values. Future studies will investigate the applicability of this Ki-67 predictive model of clinical outcomes as it pertains to surgical morbidity, extent of resection, risk of tumor recurrence, and overall survival, including in grade II and III meningiomas.

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Disclosures

The authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

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Supplementary Digital Content. Table. Supplementary Table 1: The list of selected features (n = 60) in the final model. GLCM = gray-level co-occurrence matrix; GLRLM = gray-level run-length matrix; GLSZM = gray level size zone; NGTDM = neighborhood gray tone difference matrix; LBP = local binary patterns.
